

Who leaves teaching? Predictors and the profile of teacher attrition in the United States, 2024–2025

Evidence from 6,426 school teachers followed for twelve months in linked Current Population Survey microdata

EXECUTIVE SUMMARY

- Linking individuals across the CPS rotation design, **24% of school teachers observed in 2024 were no longer teaching twelve months later**: 16.3 percentage points (pp) moved to another occupation, 6.5 pp left the labor force and 1.2 pp were unemployed.
- **Working part-time is the strongest single predictor** of leaving (+18 pp), while holding a graduate degree is the strongest protective factor (–20 pp). Exit risk is U-shaped in age, lowest around age 40.
- The **typical leaver** is female, in her early fifties or her twenties, teaching part-time, without a graduate credential, and disproportionately in preschool/kindergarten; teachers in the top predicted-risk decile face a 59% exit probability, 2.5 times the average.
- Retention policy should focus on **converting involuntary part-time positions into full-time contracts and subsidizing graduate credentials** for the least-attached segments of the teaching workforce.

24 %

of 2024 teachers had left the profession by 2025

+18 pp

extra exit risk from part-time work

–20 pp

lower exit risk with a master's degree

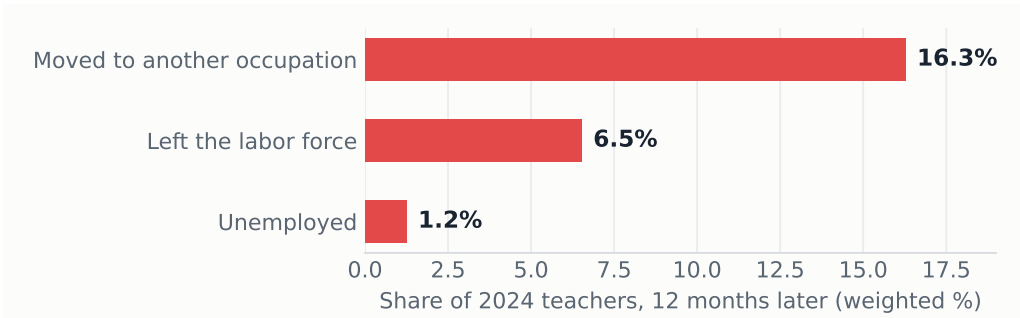
1. Data: a 12-month panel of teachers

We use official **Current Population Survey (CPS) basic monthly microdata** (U.S. Census Bureau), January 2024 to December 2025 (23 monthly files; October 2025 was not released). Exploiting the CPS 4–8–4 rotation design, each adult interviewed in 2024 is re-interviewed in the same calendar month of 2025 and can be linked by household and person identifiers, with matches validated on sex, race and age progression (Madrian–Lefgren criterion). This yields **298,586 validated individual links**, of which **6,426 were school teachers** at baseline (preschool to secondary, including special education; Census occupation codes 2300–2330). A *leaver* is a

baseline teacher who, twelve months later, is not employed as a school teacher. Reported rates use CPS final person weights.

Caveat: occupation in the returning rotation is re-coded independently, which inflates measured occupation switching; the 24% level should be read as an upper bound, while *comparisons across groups* — the object of this brief — are far less affected.

Figure 1: Most leavers move to other jobs rather than out of work. Destination twelve months later, share of 2024 teachers (weighted).

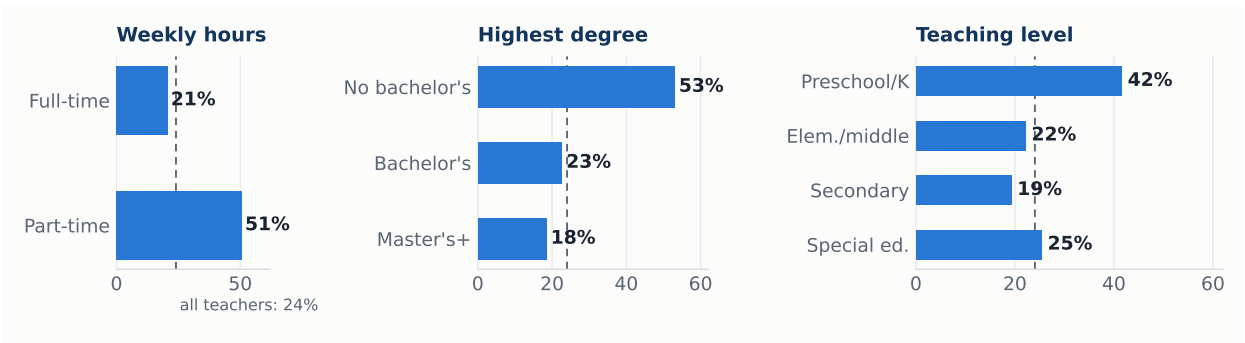


Source: authors' calculations, linked CPS basic monthly files 2024–2025.

2. Who leaves: raw differences

Attrition is heavily concentrated in three margins: **hours**, **credentials** and **teaching level**. Part-time teachers leave at more than twice the rate of full-time teachers; teachers without a bachelor's degree leave at nearly three times the rate of those with a master's; and preschool/kindergarten teachers are the most exposed level.

Figure 2: Attrition doubles without full-time hours, and falls steeply with credentials. Weighted 12-month attrition rate by baseline characteristics.



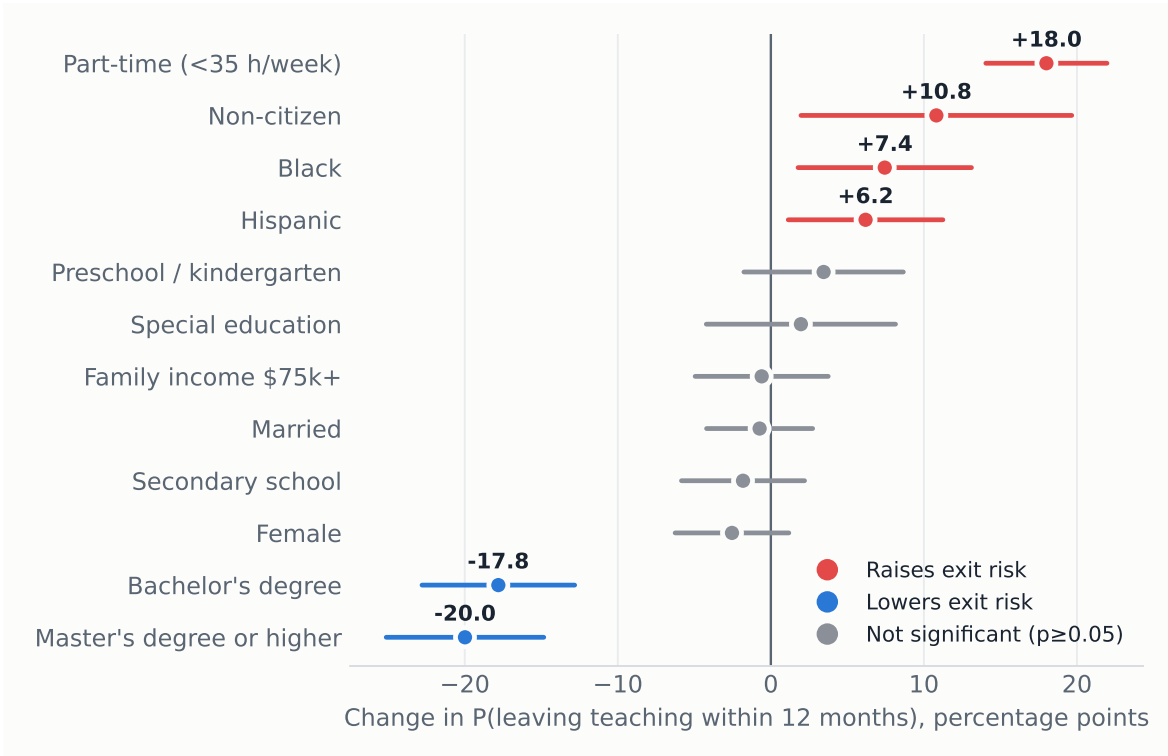
Source: authors' calculations, linked CPS 2024–2025. Dashed line: all-teacher average (24%).

3. A probit model of leaving teaching

We estimate a probit for the probability of leaving, with all covariates measured at baseline and standard errors clustered by household (Table 1). Figure 3 plots the average marginal effects. Three results stand out. First, **part-time status raises exit probability by 18pp**, the largest effect in the model — attachment to full-time hours, not demographics, is the dominant margin.

Second, **credentials protect**: a bachelor's degree lowers exit risk by 18 pp and a graduate degree by 20 pp relative to teachers without a degree, consistent with higher occupation-specific investment raising the cost of leaving. Third, **minority and non-citizen teachers are significantly more likely to leave** (+6 to +11 pp), pointing to a retention gap with direct implications for workforce diversity.

Figure 3: Hours and credentials dominate demographics. Average marginal effects on the probability of leaving teaching within 12 months, with 95% confidence intervals.



Source: probit of Table 1. Age terms shown in Figure 4.

Age enters with a robust U-shape (Figure 4): predicted risk falls through the twenties and thirties, bottoms out at 18% around age 40, and accelerates after 55 as retirement margins open — the familiar U-shaped attrition profile of the teacher-turnover literature.

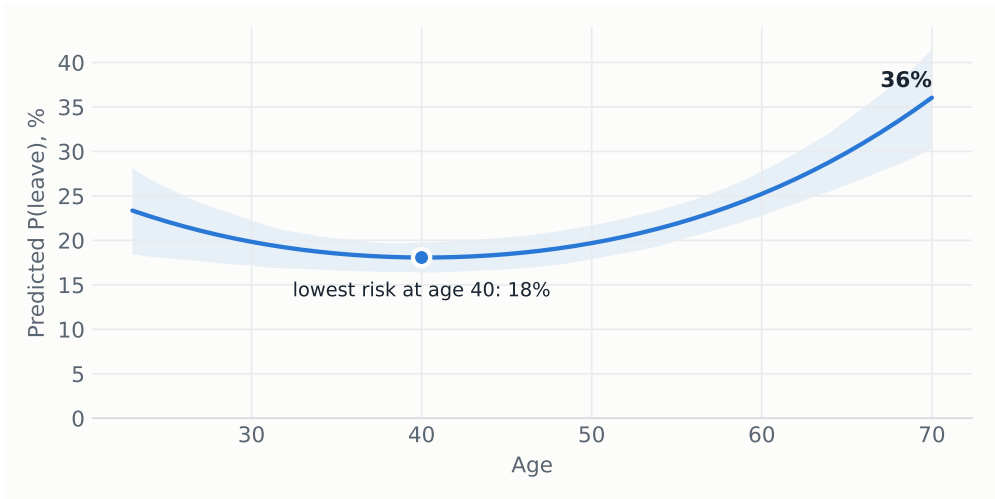
4. The profile of the typical leaver

Ranking teachers by predicted risk, the top decile — with a mean exit probability of 59%, against 23% for the average teacher — is overwhelmingly female (86%), five times more likely to work part-time (55% vs. 11%), three times less likely to hold a graduate degree (15% vs. 50%), and four times more concentrated in preschool/kindergarten (40% vs. 11%), with Hispanic and Black teachers over-represented.

5. Policy implications

The results locate teacher attrition in **weak attachment** — part-time hours, missing credentials, the early-childhood segment — rather than in broad demographics. Three levers follow directly: (i) **full-time conversion** of involuntary part-time teaching positions; (ii) **credentialing subsidies**

Figure 4: Exit risk is U-shaped in age, lowest near 40. Predicted probability of leaving by age, other covariates at sample means; shaded band: 95% CI.



Source: probit of Table 1, simulation-based confidence band.

Table 1: Probit estimates: probability of leaving teaching within 12 months.

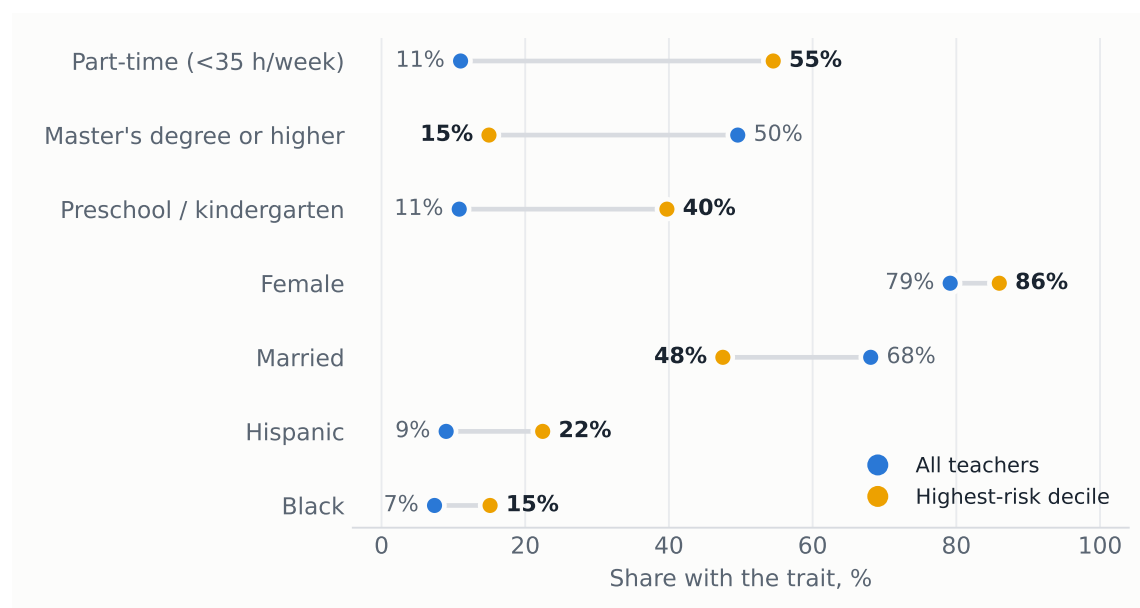
	Probit coef.	(SE)	AME (pp)
Age (years)	-0.050***	(0.014)	-1.36***
Age ² /100	0.063***	(0.014)	+1.69***
Female	-0.094	(0.070)	-2.54
Married	-0.027	(0.065)	-0.73
Black	0.275**	(0.107)	+7.44**
Hispanic	0.229**	(0.096)	+6.19**
Non-citizen	0.400**	(0.167)	+10.81**
Bachelor’s degree	-0.659***	(0.096)	-17.80***
Master’s degree or higher	-0.740***	(0.100)	-19.98***
Part-time (<35 h/week)	0.666***	(0.077)	+18.00***
Hours not reported	0.514***	(0.115)	+13.88***
Family income \$75k+	-0.022	(0.082)	-0.60
Preschool / kindergarten	0.128	(0.098)	+3.45
Secondary school	-0.067	(0.076)	-1.82
Special education	0.073	(0.117)	+1.96
Constant	0.685**	(0.315)	
Observations		6,426	
Pseudo R ²		0.104	

Cluster-robust standard errors (by household) in parentheses. AME: average marginal effect in percentage points. Reference categories: full-time, no bachelor’s degree, elementary/middle school. ****p* < 0.01, ***p* < 0.05, **p* < 0.1.

(tuition support for master’s programs) targeted at non-credentialed teachers, whose exit risk is highest; and (iii) **retention packages for preschool and kindergarten teachers**, the level where pay and conditions gaps versus other levels are largest. The significant minority and non-citizen retention gaps warrant specific monitoring, since they compound into the diversity of the future teaching workforce.

Methods in one line. Linked CPS 4–8–4 rotation panel, 2024→2025 · 6,426 teachers · probit, household-clustered SE · fully reproducible: `src/02_build_cps_panel.py` → `src/04_figures.py`.

Figure 5: The highest-risk teacher: part-time, less credentialed, early-years. Characteristics of the top predicted-risk decile vs. all teachers.



Source: predicted probabilities from the probit of Table 1.